Chi-Sheng (Michael) Chen (陳麒升)

ML Researcher & Engineer | Clinical Biosignal ML | LLM Agents & Multi-Agent Systems | Foundation Models

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Nationality: Taiwan | Based in Cambridge, MA, USA

Chi-Sheng (Michael) Chen is a machine-learning researcher and engineer working across clinical biosignal ML and LLM-agent systems. At Harvard/BIDMC he builds multimodal EMS trauma-triage pipelines (ASR → LLM-agent event extraction → structured prediction) over 4M+ patient records; his EEG-based depression-treatment model is in routine outpatient use at Taipei Veterans General Hospital. He has shipped LLM-agent systems (a US-stock research agent suite, a robotics workcell optimizer), cut LLM token usage by 82% for a quant-trading firm, and placed 3rd of 1,300+ teams at Berkeley RDI AgentX-AgentBeats 2026. Prior: Co-Founder/CTO of Neuro Industry (GCP MLOps on 10K+ EEG samples) and Digital IC R&D Engineer at MediaTek. 49 publications, 491 citations (h-index 13); reviewer for NeurIPS, ICML (Gold Reviewer) and 43 journals.

Research & Professional Experience

Harvard Medical School & Beth Israel Deaconess Medical Center — MA, USA

Research Affiliate (primary appointment), Advisor: Prof. Dr. Gabriel Brat — *Nov 2024 – Present*

- Building a real-time EMS trauma-triage pipeline (ASR → LLM-agent event extraction → structured prediction) over **4M+ patient records** using multimodal AI.
- Co-developed webcam-based open-surgical skill assessment (hand keypoint detection; CLINICCAI 2026 poster) and a capture-clock methodology for browser webcam gaze (DEMI 2026 spotlight talk).

WhaleForce (subsidiary of VICI Holdings) — Taipei, Taiwan

Fractional Token Efficiency Research Lead (Advisor) — *Aug 2026 – Present*

- Advising on LLM token-efficiency research and multi-agent architecture for AI-driven quantitative trading; token-efficiency work **cut LLM token usage by 82%**.

Omnis Labs — Remote

Co-Founder — *Sep 2024 – May 2026*

- AI-driven DeFi liquidity positioning protocol with automated market maker (AMM) strategies and backtesting systems.
- Achieved 50%–120% base fee APR on WBTC/USDC and ETH/USDC pairs (Katana on SushiSwap) using time-series model-driven AMM strategies.

Neuro Industry, Inc. — CA, USA

Researcher, Co-Founder & CTO — *Mar 2024 – Jan 2025*

- Built GCP-based MLOps pipeline, enabling training on 10K+ EEG samples.
- 3 research papers: quantum machine learning for EEG signal processing (2) and EEG-to-image diffusion (1).

Department of Computer Science, National Yang Ming Chiao Tung University — Hsinchu, Taiwan

Research Assistant, Advisor: Prof. Chun-Shu Wei — *Dec 2023 – Aug 2024*

- Developed MUSE, a multimodal contrastive learning framework for EEG-based image recognition (32 citations).
- Research on nonstationary EEG dynamics and foundation models for brain-computer interfaces.

MediaTek Inc. — Hsinchu, Taiwan

Digital IC R&D Engineer — *Dec 2021 – Oct 2023*

- Responsible for microprocessor IP development for flagship 5G smartphones' display and AMBA SoC implement.
- Hardware virtualization architecture RTL design and IP verification with UVM and SystemVerilog.

Department of Surgery, National Taiwan University Hospital — Taipei, Taiwan

Research Assistant, Advisor: Dr. Shuo-Lun Lai — *Jul 2021 – Sep 2021*

- Deep learning applications on surgery automation with YOLO-based model.
- Developed a prototype for a surgical smoke evacuation system.

Department of Psychiatry, Taipei Veterans General Hospital — Taipei, Taiwan

Research Assistant, Advisor: Prof. Dr. Cheng-Ta Li — *Sep 2019 – Jun 2021*

- Built ML models on clinical EEG from major-depressive-disorder patients to predict treatment response.

Max Planck Institute for Chemical Physics of Solids (MPI CPfS) — Dresden, Germany

Research Internship, Advisor: Dr. Alexander Komarek & Dr. Li Zhao — *Jul 2018 – Sep 2018*

- Synthesis and characterization of candidate Co-based chalcogenide superconductors.

Other: OpenAI (contractor via Mercor) — AI Trainer, RLHF preference data & model-behavior evaluation — *Mar 2025 – Oct 2025*

Agent & LLM Engineering

Berkeley RDI AgentX-AgentBeats — 3rd Place / 1,300+ teams (2026). *MadGAA*, a multi-agent OSCE medical evaluation system: role-specialized agents (examiner, patient, judge) + verifiable scoring rubric (RAG + tool use + verifier). [code] [leaderboard]

US-stock LLM agent suite (deployed). Browser agent as an explicit PLAN→LOCATE→ACT→VERIFY state machine with fault-injection recovery; layered SEC 10-K extractor; 22 research agents + 11 placebo controls with lookahead-free backtests, per-provider cost ledger, and eval dashboards.

Robotics Workcell Optimizer (live). NL → spec → robot selection → 2D layout → 3D preview via multi-LLM abstraction (Claude/OpenAI/Gemini, validation-repair loop) + OR-Tools CP-SAT; ISO 13855 safety as inviolable hard constraints. [code] [demo]

Also: *flyhypo* (connectome + PubMed + LLM → grounded, falsifiable neuron hypotheses) [code]; *paper-evidence* (verbatim-quote gate + cross-family judge to block LLM hallucination) [code]; two live NYCU RAG physics tutors (Next.js + Gemini + pgvector) [live].

Stack: LangChain/LangGraph, ReAct & function calling, multi-agent orchestration, vLLM, LoRA/QLoRA, SFT/DPO/RLHF, RAG (pgvector/FAISS/Qdrant), OR-Tools, OpenAI/Anthropic/Gemini SDKs.

Education

National Taiwan University (NTU) Taipei, Taiwan
Master of Science in Biomedical Electronics and Bioinformatics *Sep 2019 – Jun 2021*
Advisor: Prof. Dr. Cheng-Ta Li, MD & Prof. Chung-Ping Chen

- The research EEG model for TMS pattern personalized prediction is currently being applied to real outpatient patients in the Precision Depression Intervention Center (PreDIC), Taipei Veterans General Hospital.
- Master's thesis: "EEG Analysis for Prediction of Antidepressant Responses of Transcranial Magnetic Stimulation in Major Depressive Disorder Based on Attentional Convolution Time Series Prototypical Neural Network Model and Classical/Quantum Machine Learning Approaches."

National Chiao Tung University (NCTU) Hsinchu, Taiwan
Bachelor of Engineering in Electrophysics *Sep 2015 – Jun 2019*
Bachelor of Science in Interdisciplinary Science Degree Program

Skills

AI/ML: Python, PyTorch, HuggingFace, Scikit-learn, MNE, MLflow, R, MATLAB
Systems: C/C++, JavaScript, Rust, MySQL, Solidity, Git, Docker, GCP, AWS
Hardware: Verilog, SystemVerilog, UVM, HSPICE, Genus, Verdi, SG-Lint, Perl
Languages: Chinese (Native), English (Fluent)

Selected Publications

(49 total, 491 cites, h-index 13 — full list on Google Scholar)

[First author unless noted]

Chen, C.S., et al. "FreqLens: Interpretable Frequency Attribution for Time Series Forecasting" *Under review*, ICLR 2027; arXiv:2602.08768.

Chen, C.S., et al. "Large Cognition Model: Towards Pretrained EEG Foundation Model" arXiv preprint (2025), arXiv:2502.17464 — **29 citations**.

Chen, C.S., Wei, C.S. "Mind's Eye: Image Recognition by EEG via Multimodal Similarity-Keeping Contrastive Learning" arXiv preprint (2024), arXiv:2406.16910 — **32 citations**.

Chen, C.S., et al. "Exploring the Potential of Electroencephalography Signal–Based Image Generation Using Diffusion Models: Integrative Framework Combining Mixed Methods and Multimodal Analysis" JMIR Medical Informatics (2025), https://doi.org/10.2196/72027 .

Chen, C.S., et al. "Exploring the potential of QEEGNet for cross-task and cross-dataset electroencephalography encoding with quantum machine learning" Journal of Signal Processing Systems 98, 5 (2026), https://doi.org/10.1007/s11265-025-01979-2 .

Li, C.T., **Chen, C.S., et al. "Prediction of antidepressant responses to non-invasive brain stimulation using frontal electroencephalogram signals: Cross-dataset comparisons and validation"** Journal of Affective Disorders (2023), https://doi.org/10.1016/j.jad.2023.08.059 . [2nd author; model deployed at Taipei VGH PreDIC]

Invited Talks & Presentations

NVIDIA GTC 2026 (Top 8 Finalist, two posters); **MICCAI 2026** — DEMI spotlight talk and CLINICCAI poster; **IEEE ICASSP 2026** (two orals) and **2025** (oral); **IEEE Quantum Week 2025**; **DH2026** (two short papers); **QTML 2024** (poster).

Guest lectures at NYCU ECE (Wearable Chip Design), NTU CSIE (Advanced Compilers) and NYCU Maker Club; invited seminar, Dept. of Applied Physics, Tunghai University (2025–2026).

Professional Service

Conference Reviewer (2026–2027): NeurIPS, ICML (Gold Reviewer — top tier), AAAI, KDD, MICCAI, ICASSP, NLDL 2027, ACM ICAIF 2026, IEEE QAI 2026. **Program Committee:** QNRL Workshop @ IEEE WCCI 2026.

Journal Reviewer (2025–present): 43 journals incl. TPAMI, TNNLS, TASLP, JBHI, Neural Networks, npj Quantum Information, Scientific Reports; **IOP Trusted Reviewer** (2026).

Awards & Achievements

2026 NVIDIA GTC Top 8 Finalist: "GPU-Accelerated Chinese Brain-to-Text Interface Powered by NVIDIA H100 80GB."

2026 Berkeley RDI AgentX-AgentBeats: Phase 1 Healthcare Agent 3rd Place (Team MadGAA, 1,300+ teams worldwide).

2023 Certificate of the 20th National Innovation Award (Taiwan), clinical research category — for the EEG treatment-response model now deployed at Taipei VGH PreDIC.

2019 International Blockchain Olympiad (IBCOL): World Finalist competition with 100+ teams.

2017 Microsoft International Imagine Cup : 2nd place in Taiwan.

2016 International Genetically Engineered Machine Competition (iGEM): World Gold medal, Best Applied Design, Best Part Collection, Best Presentation competition with 300+ university teams.